AGENT EVALUATION SCIENCE

Synthetic Stakeholders? Validity and Stability of Open-Weight LLM Agent Panels in Participatory Model Selection

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Abstract

LLM agents are proposed as synthetic participants, but rarely tested against a real participatory deliberation. We replicate a workshop in which 12 students and advisors set the rules for selecting a university dropout early-warning model, with 12-agent panels from three small open-weight model families (10 workshops each), and apply each panel's median rules to the same candidate models. Panels recovered on average 3.5, 2.8 and 2.4 of the 7 human selection rules (Qwen, Gemma, Ministral; random panels: 1.66) and selected the human-chosen model in 4/10, 6/10 and 1/10 runs, but the family that chose it most often did so through rules the humans rejected, and panels almost never objected (2 blocking votes in 360), suggesting that small synthetic panels can rehearse but not replace stakeholder deliberation.

1 Motivation

Participatory AI asks affected people to shape algorithmic systems (Birhane et al., 2022; Delgado et al., 2023); because this is slow and costly, LLM-simulated respondents are proposed as a substitute (Argyle et al., 2023; Aher et al., 2023), despite evidence that they misrepresent and flatten the groups they stand in for (Santurkar et al., 2023; Wang et al., 2025). We evaluate synthetic panels on what participation *produces*: binding rules and the model they select.

2 Ground truth and design

In a three-hour deliberative workshop at a Danish university (Blažková et al., 2026), 9 students and 3 student advisors answered a survey before and after deliberation (Fishkin, 2009), pricing each step on two ladders in students identified per 100: adding data (course performance, enrolment, LMS activity, sensitive attributes) and lowering explainability (from a credit rule to a neural network). Their endline medians, applied to all candidate models on the 2021 cohort, selected logistic regression on credits and performance (recall 62.3% vs. 23.5% for

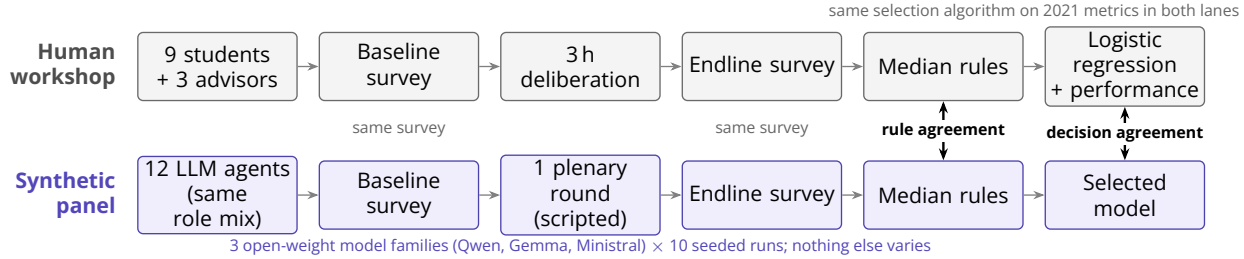


Figure 1 | Design. Synthetic panels pass through the same survey, aggregation and selection algorithm as the human workshop; the model family is the only manipulated factor.

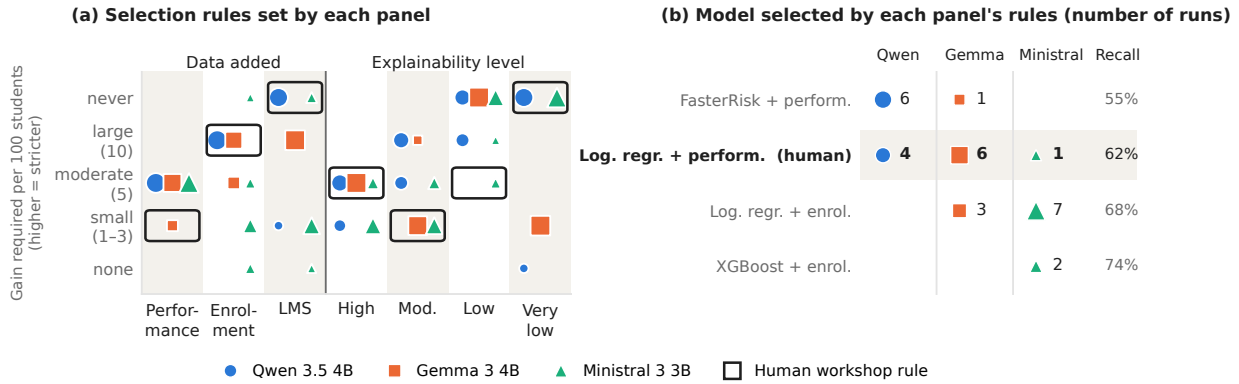


Figure 2 | (a) Selection rules set by each panel (endline medians; marker area = runs) against the human rules (boxes): above a box is stricter, below more permissive. (b) Configuration selected by each panel's rules (runs) and its 2021 recall; the human choice is shaded.

Setting	Model	Rules /7	Other /16	Human model [95% CI]	Modal	Recall gap
small, lite	Qwen 3.5 4B	3.5 (2–4)	6.4	4/10 [.17–.69]	6/10	-4.6 pp
small, lite	Gemma 3 4B	2.8 (2–3)	0.9	6/10 [.31–.83]	6/10	+1.1 pp
small, lite	Ministral 3 3B	2.4 (1–4)	5.4	1/10 [.02–.40]	7/10	+6.5 pp
large, lite	Qwen 3.5 27B	1.2 (1–2)	6.4	0/10 [-0.00–.28]	10/10	+6.1 pp
large, lite	Gemma 3 27B	1.0 (1–1)	5.4	0/10 [-0.00–.28]	10/10	+6.1 pp
large, lite	Mistral Small 3.2 24B	2.4 (2–3)	5.9	1/10 [.02–.40]	9/10	+5.5 pp
large, full	Qwen 3.5 27B	1.9 (1–3)	6.9	0/10 [-0.00–.28]	10/10	+6.1 pp
large, full	Gemma 3 27B	1.3 (1–2)	5.4	0/10 [-0.00–.28]	10/10	+6.1 pp
large, full	Mistral Small 3.2 24B	3.2 (2–4)	5.3	0/10 [-0.00–.28]	7/10	+2.0 pp
Random panel		1.66	6.38	14%	–	–

Table 1 | Rules matched of 7 (range), other assessed items of 16, runs selecting the human model [Wilson 95% CI] or the modal configuration, recall gap, items unanswered per workshop.

the current credit rule). Because the human pre-deliberation rules select the same model, rule agreement is our main validity outcome. Synthetic panels replay this process (Figure 1): 12 fictional personas with the published role and demographic mix, the verbatim survey, a scripted facilitator, one plenary round (seeded order, matched across families) and a final acceptability card. The only manipulated factor is the model family: Qwen 3.5 4B, Gemma 3 4B and Ministral 3 3B (4-bit, local, temperature 1). The scoring checklist and a random-panel reference were fixed before any model output; no personal data or transcripts were used, and 98.7% of choice items were answered (30 workshops, 2,420 calls).

3 Results

All three families recover more selection rules than random panels, but at most half of them, and none beats random panels on the 16 other assessed items (Figure 2, Table 1). The families err in different directions: Qwen panels are stricter about explainability and mostly select an integer scorecard (recall gap -4.6 pp), while Ministral panels accept enrolment data the humans priced high and select a more accurate model (+6.5 pp). Gemma panels select the human model most often (6/10), yet declare LMS data and neural networks acceptable, which the humans ruled out: the right decision from different rules. The modal configuration accounts for 6/10, 6/10 and 7/10 runs. Panels were also far more agreeable than the humans, among whom a minority blocked: 2 of 360 cards were blocking votes, and nearly all Gemma and Ministral agents reported becoming more supportive (Qwen 47 vs. 43 more critical; humans 6 vs. 5).

4 Limitations and ongoing work

Evidence covers small local models, an abridged deliberation, personas from marginal counts and one workshop, and differences are descriptive; yet model family alone changes the rules a panel writes: choosing a model means choosing an outcome. Next: full protocol, larger and closed models, mixed panels, persona detail, prompts.

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